1. Title

MITRE ATT&CK Mapping

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2. Objective

The AI SOC Copilot leverages Google's Gemini Large Language Model (LLM) to automatically map detected cybersecurity threats to the MITRE ATT&CK framework. After parsing uploaded security logs and extracting Indicators of Compromise (IOCs), the AI identifies the most appropriate MITRE ATT&CK tactic and technique, providing analysts with standardized threat intelligence for faster incident response.

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3. Workflow

```
Uploaded Log
      |
      ▼
Log Parser
      |
      ▼
IOC Extraction
      |
      ▼
Gemini Threat Analysis
      |
      ▼
MITRE ATT&CK Mapping
      |
      ▼
Risk Assessment
      |
      ▼
Incident Report
```

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4. AI-Based Mapping Process

Write:

The threat analysis module performs the following steps:

1. Receives parsed security logs.
2. Receives extracted Indicators of Compromise (IOCs).
3. Sends both as context to the Gemini LLM.
4. Gemini identifies the attack category.
5. Gemini maps the attack to the corresponding MITRE ATT&CK tactic and technique.
6. Gemini assigns a severity level, confidence score, and risk score.
7. The generated mapping is included in the incident report.

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5. Information Returned by Gemini

Create this table.

Field	Description
Attack Type	Category of detected attack
Severity	Low / Medium / High / Critical
Risk Score	Integer between 0 and 100
Confidence	AI confidence level
MITRE ATT&CK ID	Technique ID
MITRE Tactic	ATT&CK Tactic
MITRE Technique	ATT&CK Technique
Reasoning	Technical explanation
Recommended Actions	SOC remediation steps

This is directly based on your `threat_prompt.txt`.

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6. Example Mapping

Use the example that **matches your screenshots**.

Attack	MITRE ID	Tactic	Technique
PowerShell Execution	T1059.001	Execution	PowerShell

You don't need to invent more mappings because your project generates them dynamically.

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7. Example Output

Attack Type:
Advanced Persistent Threat (APT)

Severity:
Critical

Risk Score:
95

Confidence:
97

MITRE ATT&CK:
T1059.001

Tactic:
Execution

Technique:
PowerShell

This aligns with the UI screenshots you shared.

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8. Advantages

Write:

- Standardized attack classification.
- Faster threat triage.
- Easier incident investigation.
- Industry-standard ATT&CK framework.
- AI-assisted security analysis.
- Consistent incident reporting.

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9. Conclusion

The AI SOC Copilot automates MITRE ATT&CK mapping by combining parsed security logs, extracted IOCs, and Large Language Model reasoning. This approach enables consistent threat classification, accurate severity assessment, and standardized incident response aligned with modern SOC practices.